

Market Analysis — How We Pick Stocks

System overview from `server.js`, `analysis-engine.js`, `adaptiveWeights.js`. Not investment advice.

1. Two parallel paths

Path	Where	Role
Single-stock / UI analysis	API + analysis-engine	<code>calcComposite</code> / <code>calcCompositeRules</code> : 0–100 score, BUY/HOLD labels; optional ML on headline score
Universe backtest & paper trade	<code>runBacktestSimulation</code>	Five-pillar backtest composite, scheduled rebalances, risk rules

Both use fundamentals + price history; backtests can use **point-in-time (PIT)** fundamentals per rebalance when configured.

2. Fundamental layer (ranking input)

`fetchFundamentals` builds **rankingFundamentals**: Buffett-style quality, moat, ROIC, earnings quality, shareholder yield, constraint penalties, AI disruption heuristic, simplified DCF → **dcfScore** / **dcfUpside**, and **fundamentalComposite** (0–100, quality-heavy blend).

3. Five pillars — `bt_rankFullCompositeV2`

Full Composite default weights (`DEFAULT_COMPOSITE_WEIGHTS`): fundamental 60%, dcf 0%, dynamic valuation 0%, momentum 20%, value 20%.

Composite: $\text{fundamental} \times w_F + \text{dcf} \times w_D + \text{dynVal} \times w_V + \text{momentumTerm} \times w_M + \text{value} \times w_{Val}$ (+ optional momentum-quality bonus for vanilla full composite). Results sorted by score descending.

Aggressive / Turbo use different weight vectors, percentile momentum, momentum–quality blends, and vol caps.

4. Candidate filters

- Insufficient price history (~120+ bars)
- Missing fundamentals for simulation date
- Fundamental floor (strategy-dependent)
- Severe constraint penalty (unless turbo skips)
- Vol above `maxVol`
- Momentum quality below threshold when enabled

5. ML overlay (optional)

- Single-name: `calcComposite` can blend rules with RF / cluster outputs when predictions exist.
- Backtest: `applyMLAlphaRankingToCompositeRankings` or `applyMLBlendToCompositeRankings` when `skipMLRankingAdjustments` is false and env flags allow.
- Default UI backtest: rules-only unless API `mlRanking=1`.

6. Adaptive pillar weights

Non-fixed adaptive mode: `composeAdaptiveWeightsForRebalance` adjusts weights from rolling IC-style signals; conservative mode blends with regime-adjusted static weights. PIT staleness can halve fundamental/DCF pillar weights.

7. From ranked list to trades

1. **topPicks**: sector caps (applySectorLimits) on top of sorted list; adjustedTopN from regime.
2. **Sizing**: equal / inverse vol / score-weighted; regime **exposure** scales invested capital.
3. **SELLs**: not in target → turnover gate (~25% score improvement vs weakest new top name), min hold 45d for discretionary sells, overflow trim.
4. **BUYs**: underweight vs target if >\$50 drift and slots; **60d rebuy cooldown** after a SELL.
5. **Stops**: periodic checks; **REENTRY** after stop: ≥45d, price > 20DMA, spacing vs next rebalance.

8. Other strategies

Momentum only, Momentum + Value, Quality + Momentum — separate ranking functions without the full five-pillar blend.

9. Decision tree

```
START: Universe + strategy + rebalance date
├── Load prices (and PIT fundamentals if composite family)
├── Regime (if enabled) → exposure + adjustedTopN
├── Resolve pillar weights (fixed / override / adaptive / PIT halving)
├── Rank universe
│   ├── strategy branch
│   ├── filters: history, vol, fundamental floor, constraints, mom quality
│   ├── score & sort
│   └── optional ML rerank / blend
├── Sector cap → topPicks
├── Size positions (apply exposure)
├── SELL: turnover gate + min-hold + overflow
├── BUY: slots + cash + rebuy cooldown
└── Daily: stops → REENTRY rules
```

10. Single-stock labels (calcCompositeRules)

Separate rich model: Quality, Moat, Valuation, ROIC, Earnings quality, Momentum, Shareholder yield, Quality floor → penalties → grade STRONG BUY ... STRONG SELL; extreme overvaluation can force AVOID/HOLD.

11. Representative constants

Turnover score edge ~25%; min hold 45d; rebuy cooldown 60d; re-entry ≥45d and >20DMA. Exact values in server.js.